

CO2 ASSESSMENT TOOL 2:

SCENARIO-BASED ASSESSMENT

Aim

Implement Linear Search in C and analyze its suitability for an unsorted retail product list.

Problem Statement

A retail chain maintains an unsorted product list. Search products using Linear Search and answer the theory questions.

Algorithm

1. Read n products.
2. Read key.
3. Compare key with each product.
4. If found print position else not found.

Pseudocode

```
for each item
  if item==key return index
return -1
```

Source Code

```
#include <stdio.h>
#include <string.h>
int main(){
  int n,i,pos=-1; char prod[50][30],key[30];
  printf("Enter number of products: "); scanf("%d",&n);
  for(i=0;i<n;i++) scanf("%s",prod[i]);
  printf("Enter product to search: "); scanf("%s",key);
  for(i=0;i<n;i++) if(strcmp(prod[i],key)==0){pos=i;break;}
  if(pos!=-1) printf("Found at position %d",pos+1);
  else printf("Product not found");
  return 0;
}
```

Sample Input

```
5
Rice Oil Soap Sugar Milk
Sugar
```

Found at position 4

- (a) Linear Search checks each product sequentially until found.
- (b) As dataset grows, search time increases linearly. Worst case $O(n)$.
- (c) Binary Search is not beneficial for frequently updated unsorted data because sorting after every update is costly.

Best: $O(1)$
Average: $O(n)$
Worst: $O(n)$

 $O(1)$

main.c

Run

Clear

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     int n,i,pos=-1;
6     char products[50][30],key[30];
7
8     printf("Enter number of products: ");
9     scanf("%d",&n);
10
11    printf("Enter product names:\n");
12    for(i=0;i<n;i++)
13        scanf("%s",products[i]);
14
15    printf("Enter product to search: ");
16    scanf("%s",key);
17
18    for(i=0;i<n;i++){
19        if(strcmp(products[i],key)==0){
20            pos=i;
21        }
22    }
23
24    if(pos!=-1)
25        printf("Product found at position %d\n",pos);
26    else
27        printf("Product not found\n");
28 }
```

Linear Search is simple and efficient for frequently updated unsorted datasets.